
Optimizing CUDA SGEMM on the NVIDIA Tesla V100

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1 Environment

Experiments were conducted on an NVIDIA Tesla V100-SXM2-32GB (sm_70, 15.7 TFLOPS FP32 peak, 6 MB L2). The harness benchmarks square matrices $M = N = K \in \{128, 256, 512, 1024, 2048, 4096\}$, timing 50 iterations with `cudaEvent` after one warm-up run. cuBLAS SGEMM (CUBLAS_GEMM_DEFAULT_TENSOR_OP) is the reference.

2 Optimization Progression

Table 1 records GFLOPS at $N = 4096$ after each incremental change. Intermediate values are hand-recorded at each development step; the final row is the measured benchmark result.

Step	GFLOPS @ 4096	% of cuBLAS
Naive (global memory only)	2,026	14.5%
+ Shared-memory tiling	4,109	29.4%
+ 1-D block tiling (register accumulators)	7,985	57.1%
+ 2-D block tiling ($TM \times TN$ per thread)	11,195	80.0%
+ Vectorized <code>float4</code> loads/stores	11,038	78.9%
+ Warp tiling (2×2 warp grid, WNITER)	12,159	86.9%
+ Double buffering (software pipeline)	12,890	92.1%

Table 1: Cumulative GFLOPS at $N = 4096$ after each optimization. cuBLAS reference: 13,993 GFLOPS.

3 Final Kernel Design

The kernel uses a three-level tiling hierarchy plus software pipelining.

Block tile. $BM \times BN = 128 \times 128$ output tile, $BK = 16$ K-strip per outer loop iteration, executed by 128 threads (4 warps).

Warp tile. The 4 warps are arranged in a 2×2 grid. Each warp owns a $WM \times WN = 64 \times 64$ sub-tile and sweeps it in $WNITER = 4$ passes of $WSUBN = 16$ columns, so each thread holds $TM \times TN = 8 \times 4 = 32$ FP32 accumulators in registers.

Memory layout. The A tile is stored *transposed* in shared memory ($As[k*BM + m]$) so all threads in a warp read the same k -column entry—a broadcast with zero bank conflicts. The B tile is row-major for coalesced lane access. All global loads and the final C write-back use `float4` (128-bit), saturating memory bandwidth.

Synchronization and pipelining. Double buffering is realized with two shared-memory ping-pong buffers ($A_s[2]$, $B_s[2]$). At the start of each outer K iteration the *next* strip is prefetched from global memory into register staging arrays; after the BK -long FMA loop the staged data is committed to the alternate buffer. One `__syncthreads()` between commit and the next FMA loop is sufficient, hiding global load latency under compute throughput.

4 What Did Not Work

Block swizzling—reordering block indices to improve L2 reuse across the B matrix—was investigated as a follow-on. On the V100 the 6 MB L2 is large enough that the default row-major block schedule already keeps recently loaded A tiles warm across the 128-column block strip that shares them. Swizzling produced no measurable gain at any tested size and was dropped.

5 Scaling

Figure 1 shows throughput across matrix sizes. At $N = 4096$ the kernel reaches 92% of cuBLAS (12,890 vs. 13,993 GFLOPS). Performance scales well from $N = 2048$ onward (87% at $N = 2048$) but falls sharply for smaller sizes, where fixed launch overhead and low arithmetic intensity dominate.

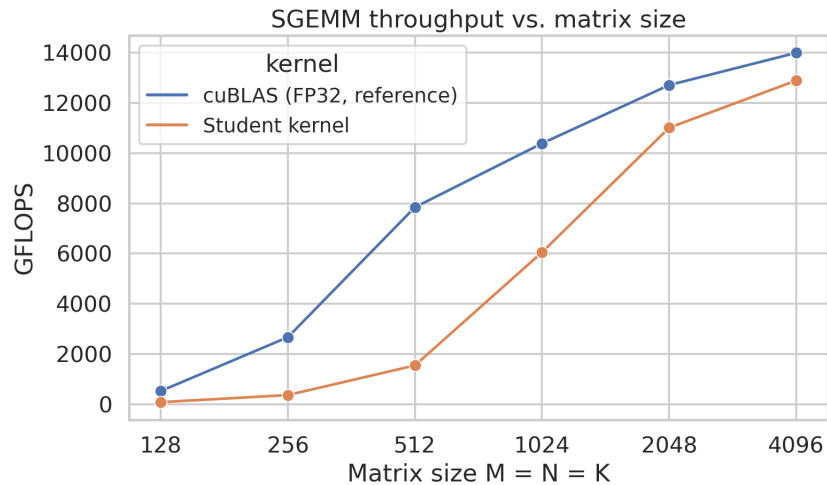


Figure 1: Student kernel vs. cuBLAS FP32 throughput across matrix sizes on Tesla V100.